

# [DHS/2024/Q4]

## 4 Name your Jupyter Notebook as:

TASK4\_<your name>\_<centre number>\_<index number>.ipynb

Your school currently keeps paper records about students who are late for school. The school's Discipline Committee wants to create a suitable database to store the data and to allow them to run reports to find out who are the repeated offenders and how many late-coming offences each class has committed. In addition, each late-coming offence results in a detention on the next weekday. The school also wants to keep track of the status of each detention to ensure that late-coming offenders learn their lesson and be punctual. You helped the school to create a web application to keep track of the late-coming and detention data. The database used for the web application has two tables: a table to store the students' information and a table to store the late-coming and detention information.

Student (StudentID, Name, Class)  
Latecoming (LatecomingID, StudentID, Latecoming\_Date, Time\_Arrived, Detention\_Date, Detention\_Completion\_Status)

Student:

- StudentID – unique student identifier, for example, T0690909J
- Name – the name of the student
- Class – the class of the student, for example, 6C33

Latecoming:

- LatecomingID – unique late-coming number, for example, 15
- StudentID – unique student identifier, for example, T0690909J
- Latecoming\_Date – the date of the late-coming offence in the form of YYYY-MM-DD
- Time\_Arrived – the time the student arrived in school, in 24-hour format
- Detention\_Date – the date of the detention in the form of YYYY-MM-DD
- Detention\_Completion\_Status – the status of the detention, for example, True if completed detention, False if otherwise

For each of the sub-tasks, add a comment statement at the beginning of the code, using the hash symbol '#' to indicate the sub-task the program code belongs to, for example:

```
In [ ]: #Task 4.1  
        Program code
```

Output:

### Task 4.1

Create an SQL file called TASK4\_1\_<your name>\_<centre number>\_<index number>.sql to show the SQL code to create the two tables, Student and Latecoming, in the database Discipline.db. Define the primary and foreign keys for each table. [2]

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## Task 4.2

The file `Latecoming.csv` stores comma-separated values for both `Student` and `Latecoming` tables.

The data in `Latecoming.csv` is given in the following order:

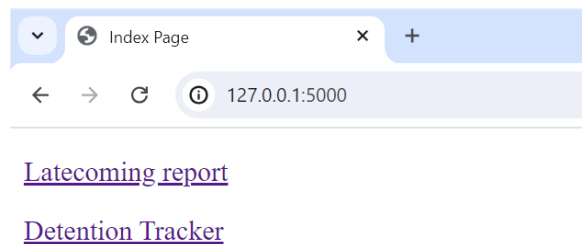
`StudentID, Name, Class, Latecoming_Date, Time_Arrived`

Write a Python program to read in the data from `Latecoming.csv` and store the data in the correct place in the database. Take note that **repeated offenders' student information should not be stored repeatedly** in the `Student` table. (Hint: Before inserting student record into `Student` table, check if the student's data has already been saved in the table before. If no, insert the student information as a new record in `Student` table. Otherwise, do not insert the student information into `Student` table again.)

In addition, the `datetime` library is built into Python and can be used to manipulate dates. See the example code shown in `Task4_datetime.py` and use the `datetime` library to generate the detention date (the **next weekday** after `Latecoming_Date`). [6]

## Task 4.3

Write a Python program and the necessary files to create a web application. The application offers the following menu options:



Save your Python program as:

`Task_4_3_<your name>_<centre number>_<index number>.py`

with any additional files/subfolders in a folder named:

`Task_4_web_<your name>_<centre number>_<index number>`

Run the web application.

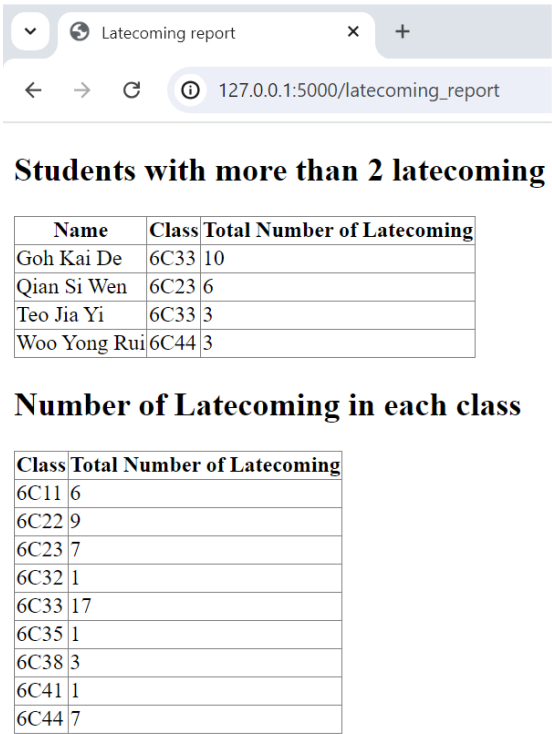
[4]

Task 4.4

Write two separate SQL queries that shows:

- the name, class and total number of latecoming for students who were late more than 2 times sorted in descending order of the total number of latecoming.
- the total number of latecoming in each class sorted in ascending order of the class name.

The results of the two queries should be shown on a web page in two tables as shown below:



This web page should be accessed from the menu option (Latecoming report) from Task 4.3.

Save your SQL queries as

Task\_4\_4\_<your name>\_<centre number>\_<index number>.sql [4]

Modify the code in your below Python program:

Task\_4\_3\_<your name>\_<centre number>\_<index number>.py

with any additional files/subfolders in the folder named:

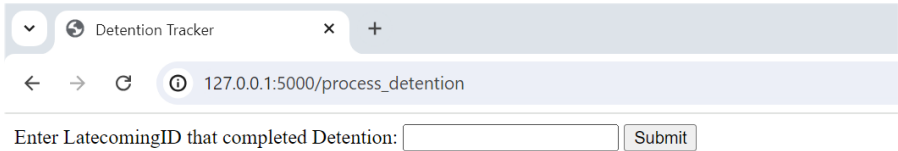
Task\_4\_web\_<your name>\_<centre number>\_<index number> [7]

Run the web application.

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Task 4.5

Modify your Python program and create the necessary file(s) to create a web page that shows the details of the detention records. There should also be a form with textbox for user to key in and submit the LatecomingID of completed detention. Upon submission of the form, the Detention Completion Status for that LatecomingID will be changed to True in the database and the change also reflected in this web page.



Detention Records:

| LatecomingID | Student Name  | Class | Latecoming Date | Detention Date | Detention Completion Status |
|--------------|---------------|-------|-----------------|----------------|-----------------------------|
| 53           | Goh Kai De    | 6C33  | 2024-07-01      | 2024-07-02     | False                       |
| 54           | Low Kai Wen   | 6C11  | 2024-07-01      | 2024-07-02     | True                        |
| 55           | Fan Yong Quan | 6C22  | 2024-07-01      | 2024-07-02     | False                       |
| 56           | Chang Hao Rui | 6C41  | 2024-07-01      | 2024-07-02     | False                       |
| 57           | Qian Si Wen   | 6C23  | 2024-07-01      | 2024-07-02     | True                        |
| 58           | Woo Yong Rui  | 6C44  | 2024-07-01      | 2024-07-02     | False                       |
| 59           | Fung Wen Kai  | 6C11  | 2024-07-02      | 2024-07-03     | False                       |
| 60           | Goh Xin Ling  | 6C33  | 2024-07-02      | 2024-07-03     | False                       |
| 61           | Goh Kai De    | 6C33  | 2024-07-02      | 2024-07-03     | False                       |
| 62           | Qian Si Wen   | 6C23  | 2024-07-02      | 2024-07-03     | False                       |
| 63           | Lim Xuan Ming | 6C22  | 2024-07-02      | 2024-07-03     | False                       |

This web page should be accessed from the menu option (Detention Tracker) from Task 4.3.

Save any additional files/subfolders in a folder named:

Task\_4\_web\_<your name>\_<centre number>\_<index number> [6]

Run the web application.

Save the webpage output as:

Task\_4\_5\_<your name>\_<centre number>\_<index number>.html [1]

Save all your files for Task 4.